

Supplemental Exercise: Reading Multivariate Plots

Day 1 — Multivariate Data & R Setup

STAT 4051 — Statistical Machine Learning I | University of Minnesota

Dataset	iris (150 flowers, 4 measurements, 3 species)
Goal	Interpret scatter plots, pair plots, densities, and correlations
Connects To	Day 1 R Lab; Chapter Notes 2 (covariance & correlation)

How to use this exercise. Each section shows a figure, explains what it is, and asks you to interpret it. Write your answers in the space provided or on separate paper. No R needed — everything you need is already on the page.

The Data

The `iris` dataset records four measurements (in cm) on 150 iris flowers from three species — 50 flowers per species.

Column	Measurement	Units
<code>Sepal.Length</code>	Length of the outer flower petal (sepal)	cm
<code>Sepal.Width</code>	Width of the sepal	cm
<code>Petal.Length</code>	Length of the inner petal	cm
<code>Petal.Width</code>	Width of the inner petal	cm
<code>Species</code>	<i>setosa</i> , <i>versicolor</i> , or <i>virginica</i>	—

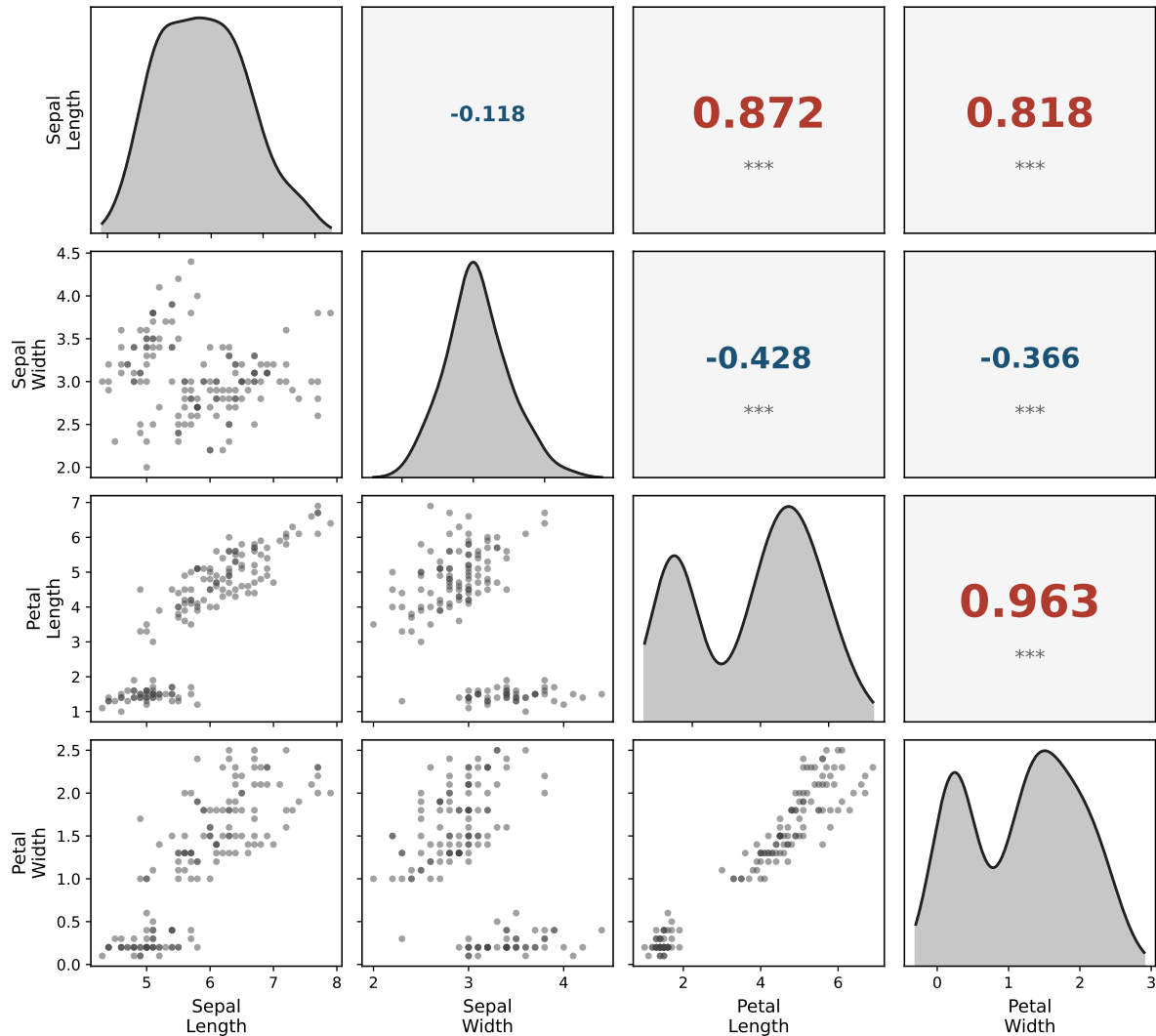
The data matrix is $\mathbf{X} \in \mathbb{R}^{150 \times 4}$ (ignoring the species label). Each row is one flower; each column is one feature.

Figure 1: The Pair Plot (no grouping)

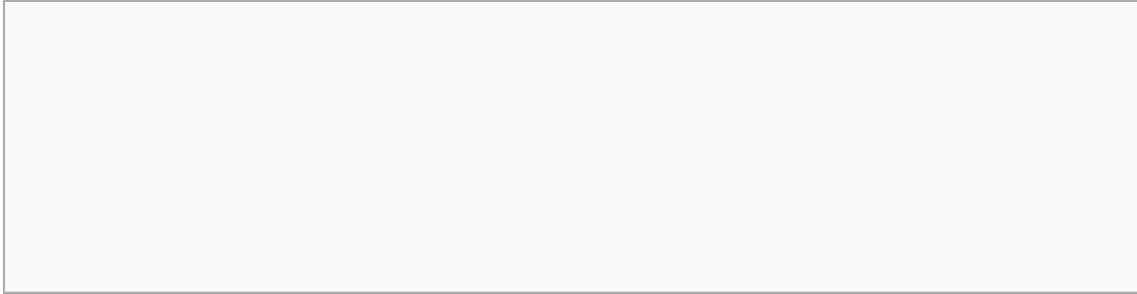
A **pair plot** (also called a scatter matrix, or the output of `ggpairs()` in R) shows every pairwise combination of variables in one grid. For $p = 4$ variables there are $\binom{4}{2} = 6$ unique pairs, so the grid is $4 \times 4 = 16$ panels. Here is how to read it:

- **Diagonal panels** (top-left to bottom-right): each shows the marginal distribution of one variable, as a smoothed density curve. The x -axis spans the range of that variable; the y -axis is density.
- **Lower-triangle panels** (below the diagonal): each shows a scatter plot of the variable on the y -axis vs. the variable on the x -axis. The column name labels the x -axis; the row name labels the y -axis.
- **Upper-triangle panels** (above the diagonal): each shows a single number — the *Pearson correlation coefficient* r between that pair of variables. The font size is proportional to $|r|$: a large bold number means a strong relationship, a small number means a weak one. Red indicates positive correlation; blue indicates negative. Asterisks below the number indicate statistical significance: * means $p < 0.05$, ** means $p < 0.01$, *** means $p < 0.001$.

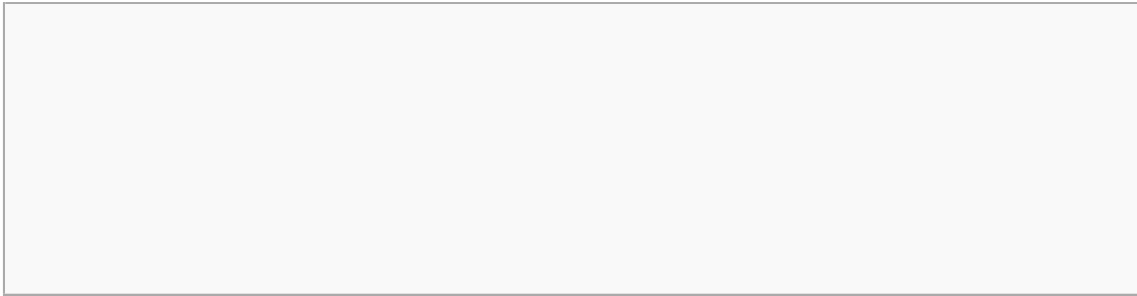
Questions — Figure 1

Figure 1 — Pair Plot: all 150 observations (no grouping)**Figure 1:** Pair plot of the four *iris* measurements. All 150 flowers are treated as a single group.

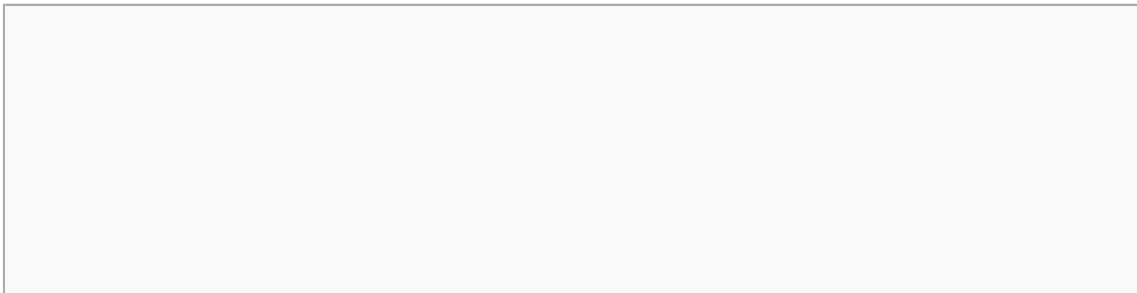
1. **Diagonal panels.** Look at the density curve for *Petal Length* (third diagonal panel). Describe its shape. Is it approximately bell-shaped (unimodal)? Does it look like it might be hiding more than one group of flowers? Contrast it with the density for *Sepal Width* (second diagonal panel).



2. **Reading a scatter panel.** Find the scatter plot where *Petal Length* is on the x -axis and *Petal Width* is on the y -axis. (Hint: it is in the lower triangle; Petal Length is column 3, Petal Width is row 4.)
- (a) Describe the overall shape of the scatter. Is it roughly linear?
 - (b) Are there any obvious gaps or clusters in the point cloud?
 - (c) Without looking at the upper triangle yet: would you guess the correlation is positive, negative, or near zero?



3. **Reading the upper triangle.** Now look at the upper-triangle panel in row 3, column 4 (or equivalently, row 4, column 3 in the lower triangle). You should see a large bold red number.
- (a) What is the value of r ? Does its sign match your prediction above?
 - (b) The font is large because $|r|$ is large. Compare the font size to the upper-triangle panel for *Sepal Length* vs. *Sepal Width*. Which pair has a weaker linear relationship?
 - (c) The number of asterisks is `***`. In plain English, what does this mean?



4. **Spot-check symmetry.** The pair plot is symmetric about its diagonal. Find the scatter plot of Sepal Length (x) vs. Sepal Width (y) and the scatter plot of Sepal Width (x) vs. Sepal Length (y). Are these the same plot or mirror images? Does the correlation in the upper triangle change depending on which panel you look at? Why not?

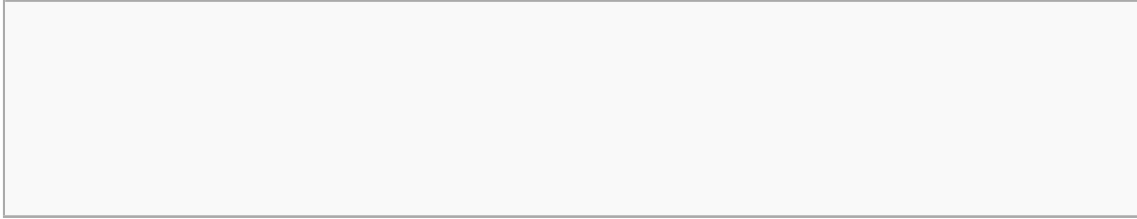


Figure 2: The Pair Plot (coloured by species)

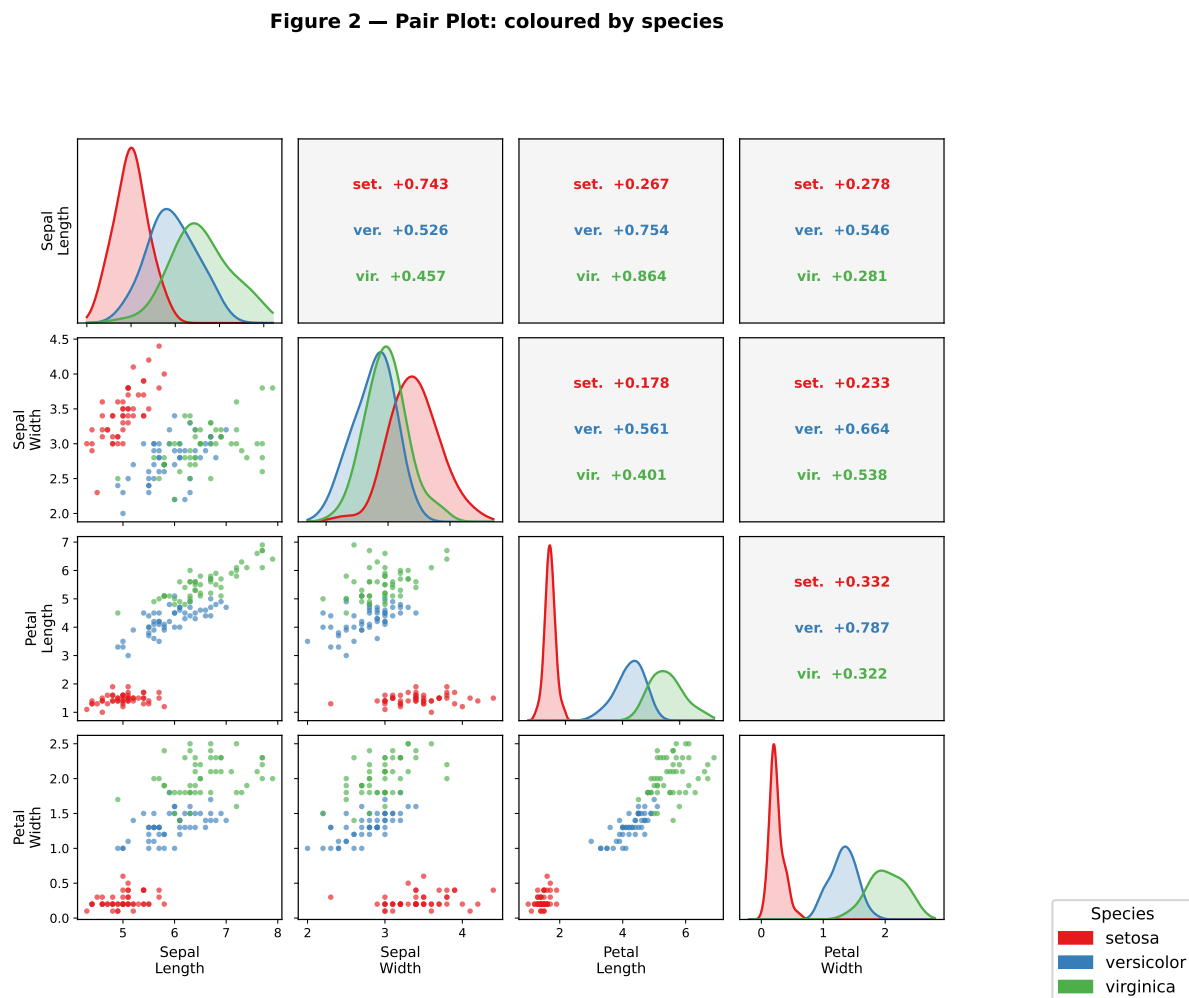
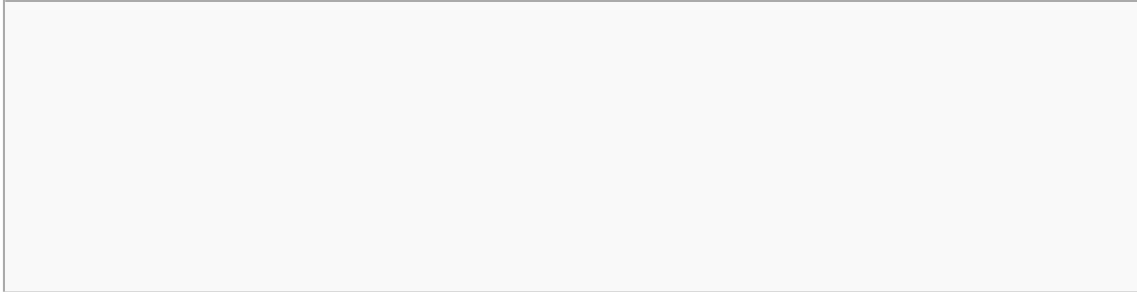


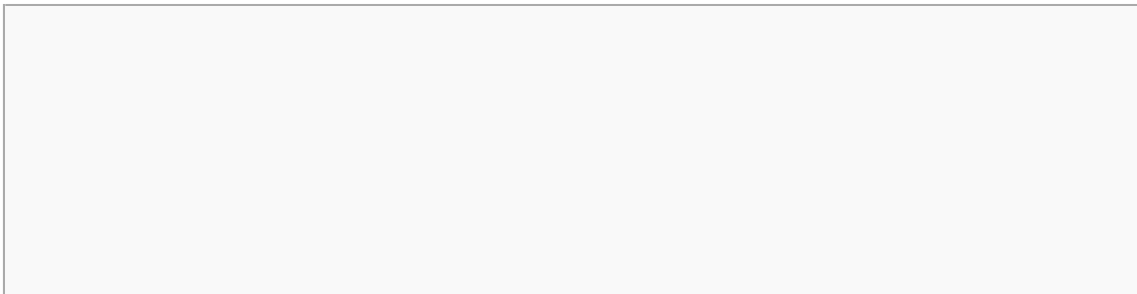
Figure 2: The same pair plot as Figure 1, but each flower is coloured by species: **setosa** (red), **versicolor** (blue), **virginica** (green). Upper-triangle panels now show the per-species correlation coefficients rather than a single overall value.

Questions — Figure 2

5. **Diagonal panels with colour.** Look at the Petal Length diagonal panel again. Now that flowers are coloured by species, how does the density picture change?
- (a) How many peaks (modes) do you see, and which species corresponds to each?
 - (b) *Setosa* has the smallest petal lengths. Are the three species well-separated on this variable, or do they overlap heavily?
 - (c) Compare this to the Sepal Width diagonal panel. Which variable does a better job of separating the three species?



6. **Scatter panels with colour.** Look at the scatter plot for Petal Length vs. Petal Width (lower triangle).
- (a) Do the three species form visually distinct clusters?
 - (b) Which two species are hardest to separate on this pair of variables?
 - (c) In Figure 1 the overall point cloud looked like one elongated blob. In Figure 2 you can see it is actually three separate clusters. Why did the overall cloud appear so elongated in Figure 1?



7. **Reading per-species correlations in the upper triangle.** Find the upper-triangle panel for Petal Length vs. Petal Width (row 3, col 4, or row 4, col 3). It shows three correlation values in three colours.
- (a) Write down the three per-species correlations. Are they all positive?
 - (b) Compare them to the overall r you found in Question 3(a). Is the overall value larger or smaller than the within-species values? Why might that be?

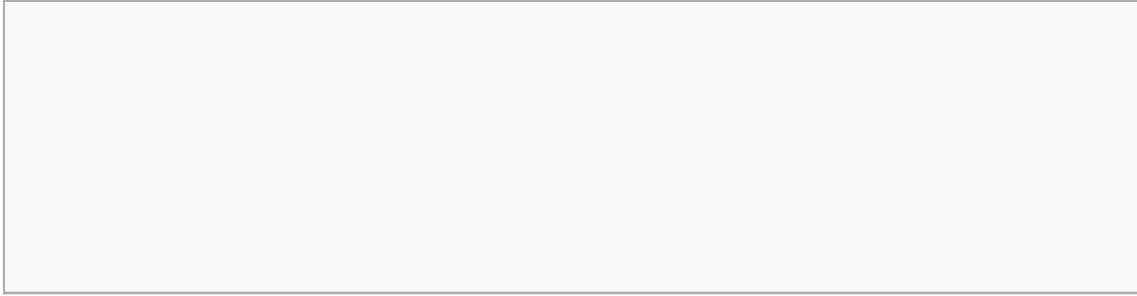


Figure 3: Focused Scatter — Petal Length vs. Petal Width

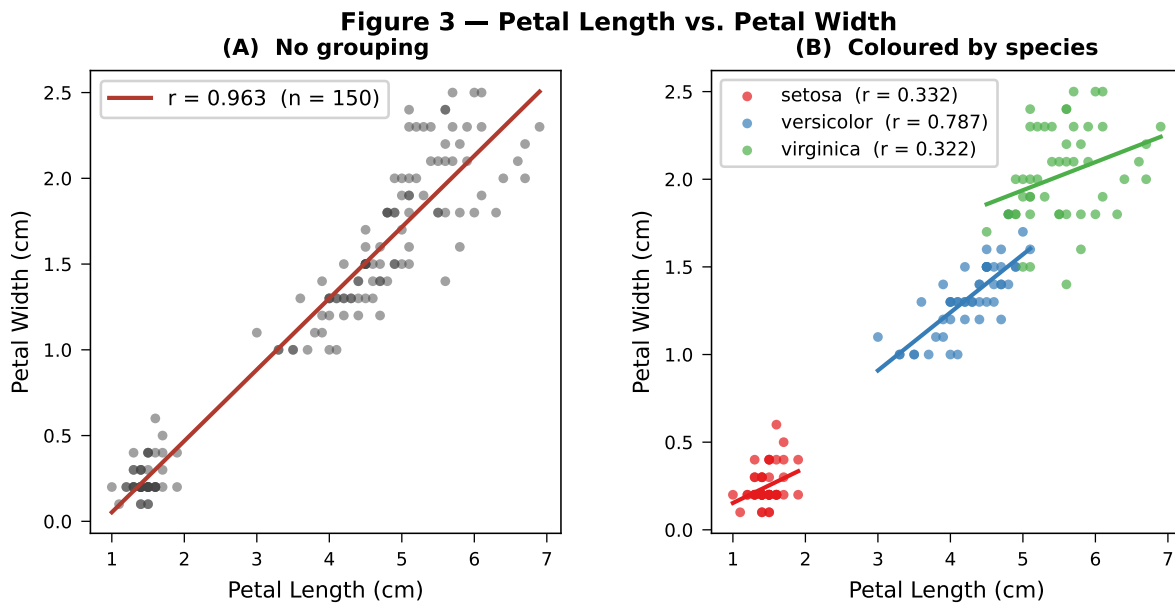


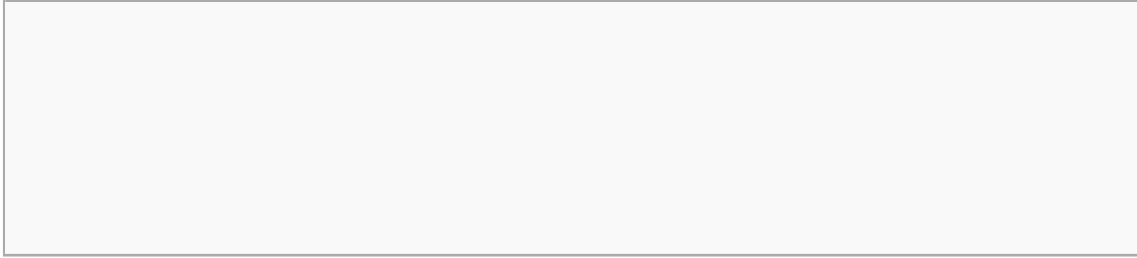
Figure 3: Petal Length vs. Petal Width. Panel (A) shows all 150 flowers in grey with a single regression line. Panel (B) shows the same data coloured by species, with one regression line per species.

A regression line is the best-fit straight line through the scatter; its slope and sign match the sign of r .

Questions — Figure 3

8. Overall regression line (Panel A).

- The correlation is shown in the legend. Using the interpretation $r = \cos \theta$ (angle between centered column vectors), roughly what angle in degrees does this correspond to? Use the fact that $\cos(15^\circ) \approx 0.97$.
- Is the scatter tight around the regression line, or is there a lot of spread? What does this say about how well Petal Length predicts Petal Width?



9. **Per-species regression lines (Panel B).**

- (a) All three per-species lines have positive slopes. Does the overall line in Panel A also have a positive slope? Is this consistent?
- (b) Looking at the legend, which species has the strongest within-species correlation?
- (c) The three species occupy different regions of the plot. *Setosa* flowers have short, narrow petals; *virginica* flowers have long, wide petals. Even if the within-species correlation were zero, would the overall r still be positive? Explain in one or two sentences.

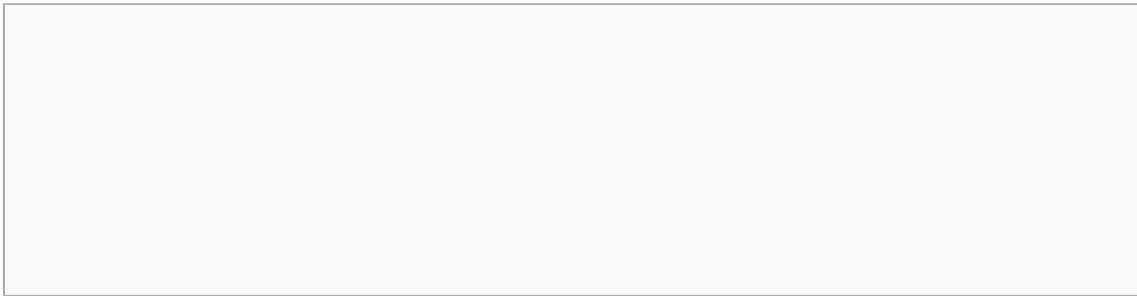
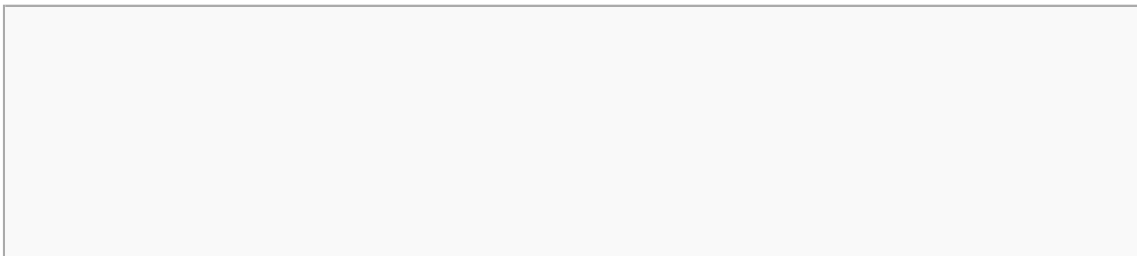


Figure 4: Sepal Length vs. Sepal Width — Simpson's Paradox

Questions — Figure 4

10. **The sign flip.**

- (a) What is the sign of the overall r in Panel (C)? What is the sign of each per-species r in Panel (D)?
- (b) These signs are *opposite*. Describe in your own words what the overall trend says vs. what the within-species trend says.



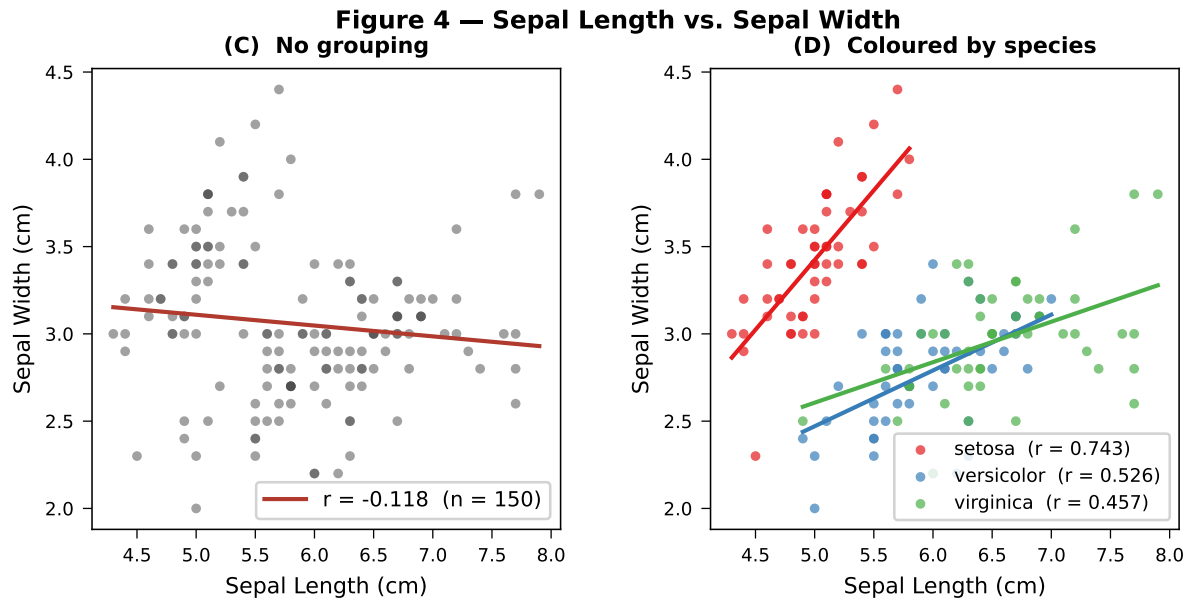
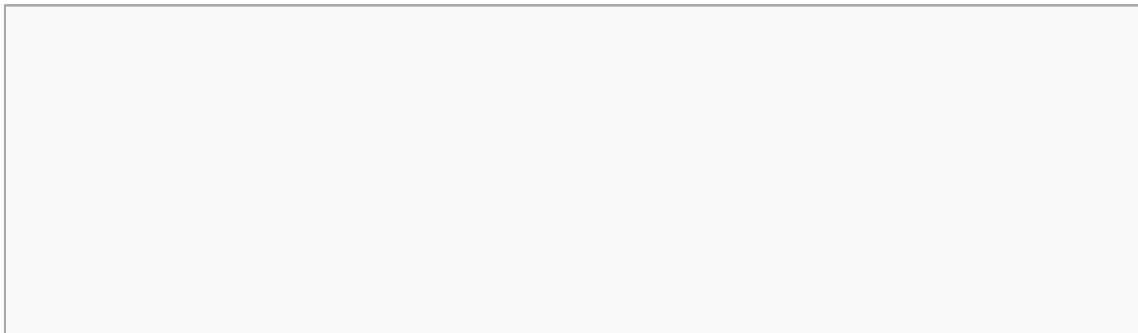


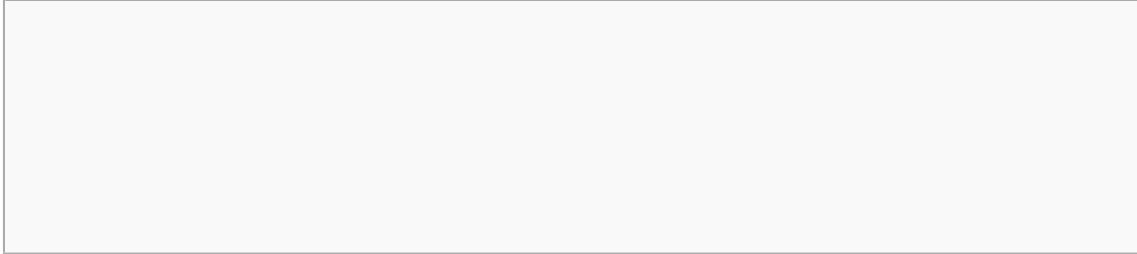
Figure 4: Sepal Length vs. Sepal Width. Panel (C) ignores species; Panel (D) reveals the per-species structure. Per-species correlations are shown in the legend of Panel (D).

11. **Why does the sign flip?** Look at Panel (D). Notice that *setosa* (red) flowers are in the upper-left region (short sepal, wide sepal), while *virginica* (green) flowers are in the lower-right (long sepal, narrower sepal relative to its length).
- When you ignore species and pool all 150 points, which direction does the point cloud lean? Trace the overall trend mentally.
 - This phenomenon — where a trend within subgroups is reversed or masked when groups are pooled — is called **Simpson's Paradox**. Describe in one sentence why this paradox occurs here.
 - Why is this important for multivariate data analysis? What would you be missing if you only ever looked at ungrouped scatter plots?



12. **Connection to the covariance matrix.** Recall from Chapter Notes 1 that the sample covariance between Sepal Length and Sepal Width is $s_{12} = -0.042$.

- (a) Does the sign of s_{12} match the sign of r in Panel (C)? It should — explain why.
- (b) If you computed the covariance matrix *separately* for each species, would the (1, 2) entry be positive or negative for most species? What does Figure 4 suggest?



Summary: A Framework for Reading Pair Plots

After working through these figures, here is a checklist you can apply to any pair plot you encounter.

1. **Diagonal first.** Look at each density. Is it unimodal or does it have multiple peaks? Multiple peaks often signal hidden subgroups.
2. **Upper triangle for a quick overview.** Scan the correlation values. Find the largest (in absolute value) and smallest. Large font = strong linear relationship; small font = weak. Red = positive; blue = negative.
3. **Lower triangle for the shape.** The scatter plot tells you whether the relationship is actually linear, curved, fan-shaped, or composed of clusters that a single number cannot describe.
4. **Add colour.** If you have a grouping variable, colour by it. Per-group correlations often differ from the pooled value — sometimes dramatically (Simpson's Paradox).
5. **Ask: does pooling make sense?** If the groups occupy very different regions of the plot, the pooled correlation may be dominated by *between*-group differences rather than *within*-group linear relationships.

This framework applies directly to the covariance matrix **S** and correlation matrix **R** you will compute in Chapter Notes 2 and R Lab 2.